Simulation of Forage Production Systems with a Weather-based Model

I. Model Description

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ABSTRACT

In cool-temperate climates, red clover/grass-mixtures play an important role in fodder production. Especially within the context of increasing agricultural extensification and in view of increasing costs of nitrogen fertilizers red clover will further gain more importance. The yields and fodder qualities of clover/grass-mixtures are variable and often small. The objectives of this study were (1) to proove the utility of a weather-based simulation model for the prediction of yield and quality of clover/grass-mixtures and (2) to assess the performance of the model for such crops.

Sensitivity tests were performed with the model to investigate the importance of various meteorological and physiological parameters on biomass accumulation and quality of herbage in a simulated spring growth.

Keywords: Red clover/grass-mixtures, dynamic model, yield, quality.

INTRODUCTION

The traditional way in fodder production research is to get evidence from the conduction of experimental field trials. The numerous they are conducted the more they can cover the natural variability of environmental conditions. In view of increasing costs in research and shrinking resources in the future, the conduction of sufficient and informative field trials will become more difficult. Models enable users to simulate yield and quality of fodder stands under a great

variability of weather and site conditions. By this way research and advisory work can achieve a maximum efficiency.

MATERIALS AND METHODS

Model Development

In order to describe environment x management interactions for grassland production systems the work started with a dynamic weather-based model for dry matter production (Angus 1980; Torssell and Kornher 1983). According to the dynamic principle a model for prediction of nutritive value of forage on the basis of daily weather datas was developed (Petersen-Fredrich 1987; Kornher *et al* 1992). Model development continued with the assembly of both models for a simultaneously calculation of production and change of forage quality of grass dominated swards (Konher and Nyman 1992). Recent research work resultet in the incorporation of a model to calculate the content of nonstructural carbohydrates in forage grasses (Wulfes *et al* 1995; Nyman *et al* 1996).

Model Design

GROWTH Submodel

$$\mathbf{W_t} = W_{(t-1)} + (WR_{(t-1)} \cdot RS_t \cdot AGE_t \cdot GI_t) \qquad (1)$$

where W_t = dry matter yield at day t [$g \cdot m^{-2}$],

$WR_{(t-1)}$ = growth rate of sward [$g \cdot m^{-2}$],

RS = initial value of the relative growth rate at the beginning of the growth period [%],

AGE = leaf area dependent index,

GI_t = environmental change rate [%], as described below.

Sward specific Parameters

$$\mathbf{AGE} = 1 / (1 + (LAI / LAI_{hm})^{a}) \qquad \text{with } 0 < \text{AGE} \leq 1 \qquad (2)$$

where LAI = leaf area index [$dm^2 \cdot dm^{-2}$], LAI_{hm} = 50%-value of maximal LAI [$dm^2 \cdot dm^{-2}$], a = constant defining the curvature of the AGE-function.

$$\text{LAI} = (W_t / b)^{0.5} \qquad (3)$$

where W_t = assimilating dry matter yield [g / m^2], b = constant dependent on species.

$$\text{RA} = RS \cdot AGE \qquad (4)$$

where RS = initial value of the relative growth rate at the beginning of the growth period [%], AGE = leaf area dependent index, see (2).

Environment affected Parameters

$$\text{GI} = TI \cdot RI \cdot WI \qquad (5)$$

where TI = temperature index, RI = radiation index, WI = water index.

$$\text{RI} = (1 - e^{(c \cdot R/(R_{opt} \cdot LF))}) / (1 - e^{(-c)}) \qquad \text{with } 0 < \text{RI} \leq 1 \qquad (6)$$

where R = daily sum of incoming short wave radiation [$J \cdot cm^{-2}$], R_{opt} = daily sum of incoming short wave radiation at full radiation supply [$J \cdot cm^{-2}$], c = constant defining the curvature of the RI-function.

$$\text{LF} = (1 - e^{(d \cdot LAI / LAI_m)}) / (1 - e^{(-d)}) \qquad \text{with } 0 < \text{LF} \leq 1 \qquad (7)$$

LF takes into consideration, that growth of plants cannot be increased over a maximum radiation, which depends on actual leaf area index of sward. LAI = leaf area index [dm^2/dm^{-2}], LAI_m = treshold of leaf area index above which maximum radiation is not affected by leaf area index [$dm^2 \cdot dm^{-2}$], d = constant defining the curvature of the LF-function.

$$\text{TI} = (1 - ((2 \cdot |T - T_m|) / (T_m - T_0))^{e}) / 2 \qquad \text{with } 0 < \text{TI} \leq 1 \qquad (8a)$$

$$TI = ((2 \cdot (1 - |T - T_m|) / (T_m - T_0))^f) / 2 \qquad \text{with } 1 \geq TI > 0 \qquad (b)$$

where T = daily mean temperature at a height of 2 m, T_m = daily mean temperature of maximum growth, T_0 = minimum daily mean temperature, when growth starts, e rsp. f = constants defining the curvatures of the functions. TI-function (8a) counts for the range between the inflection points of the functions $(T_m - T_0) / 2$ and $T_m + (T_m - T_0) / 2$. TI-function (8b) counts for the upper and lower ranges of the functions from the inflection points to the thresholds T_0 respectively $T_m + (T_m - T_0)$.

$$WI = SW_a / SW_o \qquad \text{with } 0 < WI \leq 1 \qquad (9)$$

where SW_a = actual plant available soil water amount [mm] in the root zone of the sward, SW_o = fraction of soil water capacity when the deficit of soil water starts to limit plant growth.

$$SW_a(t) = SW_a(t-1) + P(t) - TD(t) - (E_p \cdot WI(t-1) \cdot ((9 + LAI(t) \cdot 7) / 30) \qquad (10)$$

where (P_t) = actual amount of precipitation [mm] at day t, (TD) = actual amount of seepage water [mm], (E_p) = potential evapotranspiration [mm] as proposed by Penman (1963), LAI = leaf area index of sward, see (3).

TABLE 1

QUALITY Submodel

It is assumed that quality changes of fodder plants occure gradually during a growing period. Starting from one level and approaching another level, the changes generally follow a function of switch-on or switch-off type. The level and the change rates are influenced both genetically and by environmental conditions, including management factors. In the model quality changes are explained by the meteorological factors air temperature, incoming short-wave radiation, photo period and plant available soil water.

Increased temperature is assumed to enhance quality changes in swards, so the impact is expressed by a temperature change rate TCHR:

$$TCHR_d = 1 - e^{-tkq\,(t_d - tthrq)} \qquad \text{with } 0 \leq TCHR \leq 1 \qquad (11)$$

where t_d = daily mean temperature [°C], *tkq* = temperature constant, *tthrq* = temperature treshold [°C].

Increased radiation is assumed to decelarate ageing of plants, and ist effect is expressed by a radiation change rate RCHR:

$$RCHR_d = 1 - (1 - e^{-rkq\,(r_d - rthrq)}) \qquad \text{with } 0 \leq RCHR_d \leq 1 \qquad (12)$$

where r_d = daily sum of incoming short wave radiation [J·cm^{-2}], *rkq* = radiation constant, *rthrq* = radiation threshold [J·cm^{-2}].

The impact of an extended photo period is assumed to accelerate quality changes and expressed by a photo period change rate PCHR:

$$PCHR_d = 1 - e^{-pk\cdot(pp_d - pthr)} \qquad \text{with } 0 \leq PCHR_d \leq 1 \qquad (13)$$

where pp_d = photo period as a function of latitude and time of the year [hours], *pk* = photo periodic constant, *pthr* = photo periodic threshold [hours].

$$MICHR_d = 1 / (SW_o / WI_d) \qquad \text{with } 0 < MICHR_d \leq 1 \qquad (14)$$

where SW_o = soil moisture constant as a fraction of the *WI* (water index, see (4)) from the GROWTH submodel.

The four change rates (11) - (14) are multiplied into a product, which can be desribed as an environmental change rate $PECHR_d$.

$$PECHR_d = TCHR_d \cdot RCHR_d \cdot PCHR_d \cdot MICHR_d \qquad (15)$$

In the model two different functions are integrated, which describe the approximately sigmoid change of quality parameters during growth of fodder stands. These functions are switch on- (increasing parameters like ADF and crude fibre, Q_i) rsp. switch off-functions (decreasing parameters like crude protein and digestibility, Q_d) as described by Thornley (1976).

$$Q_i = Q_{min} + ((SPECHR / K_1)^{K_2} / (1 + (SPECHR / K_1)^{K_2})) \cdot (Q_{max} - Q_{min}) \qquad (16)$$

$$Q_d = Q_{min} + (1 / (1 + (SPECHR / K_1)^{k_2})) \cdot (Q_{max} - Q_{min}) \quad (17)$$

where Q_{min} rsp. Q_{max} = minimum rsp. maximum value of quality parameter, SPECHR = sum of daily environmental change rate, K_1 = defines the position of the inflection point of function, K_2 = defines the curvature of function.

Adaption of growth start in the model in spring is dependent on mean temperature in a specific period (e.g. t>5°C over a ten day-period), so different claims of temperature of genotypes are taken into consideration.

TABLE 2

Sensitivity Tests

Field Experiments

Two experiments were conducted in 1992 at the National Research Station Schaedtbek (??°??'S, ??°??'E) and at the University Research Station Karkendamm (??°??'S, ??°??'E), North Germany. The soil characteristics were as follows: Schaedtbek, Karkendamm. The experimental layout was described in detail by Wachendorf (1995), but is briefly reviewed here. In autumn swards of Lolium perenne cv. Fennema and Lolium multiflorum cv. Lema were established with red clover cv. Lero as well as pure grass swards. At both sides an identical experimental design with three replicates was used. In the present paper data of both 3 cut and 4 cut system were integrated to get a wide range of different growth characteristics. While the clover and clover/grass swards received no nitrogen fertilizer, the pure grass swards were intensively fertilized in order to realize the maximum yield potential of the experimental site. They received annual amounts of 330 (3 cut p.a.) respectively 480 kg N/ha (4 cuts p.a.) in equal dressings to each cut.

Sampling and Chemical Analysis

Plots were sampled by cutting 50 mm above ground level in weekly intervals over the whole growing period from april until october. Samples were weight immediatelly and sub-samples of

with a 1 mm sieve near-infrared spectras were taken with a NIRSystem 5500 spectrometer. By using a calibration software (ISI 1991) about 100 samples p.a. were selected by spectral characteristics to be analysed in laboratory. Another 90 samples were selected according to the Mahalanobis distance to get independent validation samples representing the total variation of plant material predicted afterwards by NIRS equations (Table 1). In selected samples nitrogen concentration (Kjeldahl digestion followed by colorimetric ammonia determination) and in vitro organic matter digestibilities were determined according to the technique of Friedel (1990). The latter procedure involves a preliminary 24-h incubation with pepsin/HCL, followed by 10-min heating in a boiling (100°C) waterbath and a final 48-h incubation with a commercial cellulase/xylanase preparation (UNIZYM F, IFZ, Berlin, Germany). This method was calibrated against a set of 206 in vivo trials with a wide source of herbages and roughages including 121 fresh crops (Friedel, 1990).

TABLE 3

RESULTS AND DISCUSSION

CONCLUSIONS

Especially within the context of increasing agricultural extensification and in view of increasing costs of nitrogen fertilizers a reconsideration of red clover is warranted and the development of suitable models to predict yield and quality of red clover based swards are necessary.

ACKNOWLEDGEMENTS

REFERENCES

Angus J F, Kornher A and Torssell B W R 1980 *A Systems approach to estimation of Swedish ley production.* Progress report 1979/80. Rapport 85. Swedish University of Agricultural Sciences, Department of Plant Husbandry (in swedish).

Friedel K 1990 Prediction of rough forage energy value by means of an cellulase technique. *Wiss. Z. Univ. Rostock, N-Reihe* **39/8**, 78-86 (in german).

Infrasoft International 1991 *Routine operation and calibration software for near infrasoft instruments.* Perstorp Analytical Inc., Silver Spring, MD 20904.

Kornher, A and Nyman P 1992 A model for prediction of growth and quality change of grass swards. *14th General Meeting of the European Grassland Federation, Lahti/Finland,* 378-382.

Kornher A, Nyman P and Taube F 1991 A model to calculate quality and change of quality of grass dominated swards by means of weather datas. *Das wirtschaftseigene Futter* **37** 232-248 (in german).

Nyman P, Wulfes R and Kornher A 1996 Modelling nonstructural carbohydrates in forage grasses with weather data. *Agricultural systems* (submitted).

Penman H L 1963 Vegetation and hydrology. Chapter 4: Evaporation, transpiration and evapotranspiration. *Technical Communication No. 53.* Commonwealth Bureau of Soils, Harpenden.

Petersen-Fredrich, E C 1987 *Quantitative description of yield formation and quality change of grassland swards by means of dynamic computer-based models.* Ph.D. Christian-Albrechts-University, Kiel (in german).

Thornley, J H M 1976 *Mathematical models in plant physiology: A quantitative approach to problems in plant and crop physiology.* Academic Press, London.

Torsell, B W R and Kornher A 1983 Validation of a yield prediction model for temporary grasslands. *Swedish Journal of Agricultural Research* **13** 125-135.

Wachendorf, M 1995 *Yield formation and changes of quality during growth of red clover and clover-grass mixtures as affected by the cutting frequency, level of nitrogen fertilizer and grass species.* Ph.D. Christian-Albrechts-University, Kiel (in german).

Wulfes R, Nyman P und Kornher A 1996 A model to calculate the contents of water soluble carbohydrates by means of weather datas. *Mitteilungen der Gesellschaft für Pflanzenbauwissenschaften* **8**, 195-198 (in german).

TABLE 1

Parameters of the GROWTH submodel

no	symbol	description	unit	function
1	a	exponential constant of the AGE-function	-	(2)
2	b	exponential constant of the LAI-function	-	(3)
3	c	exponential constant of the RI-function	-	(6)
4	d	exponential constant of the LF-function	-	(7)
5	e	exponential constant of the TI-function	-	(8a)
6	E_p	Potential evaporation	mm	(10)
7	f	exponential constant of the TI-function	-	(8b)
8	LAI_{hm}	50%-value of maximal leaf area index	$dm^2 \cdot dm^{-2}$	(2)
9	LAI_m	treshold of leaf area index	$dm^2 \cdot dm^{-2}$	(7)
10	P	actual amount of precipitation	mm	(10)
11	R	daily sum of incoming short wave radiation	$J \cdot cm^{-2}$	(6)
12	R_{opt}	daily sum of incoming short wave radiation at full radiation supply	$J \cdot cm^{-2}$	(6)
13	R_s	initial value of the relative growth rate at the beginning of the growth period	%	(4)
14	SW_a	actual plant available soil water amount in the root zone of the sward	mm	(9)
15	SW_o	fraction of soil water capacity when the deficit of soil water starts to limit plant growth	%	(9)
16	TD	actual amount of seepage water	mm	(10)
17	T_m	daily mean temperature of maximum growth	°C	(8a u. b)
18	T_o	minimum daily mean temperature, when growth starts	°C	(8a u. b)
19	WR	growth rate of sward	$g \cdot m^{-2}$	(1)

TABLE 2

Parameters of the QUALITY submodel

no	symbol	description	unit	function
1	t_d	daily mean temperature	°C	(11)
2	tkq	temperature constant	-	(11)
3	tthrq	temperature treshold	°C	(11)
4	r_d	daily sum of incoming short wave radiation	$J \cdot cm^{-2}$	(13)
5	rkq	radiation constant		(13)
6	rthrq	radiation threshold	$J \cdot cm^{-2}$	(13)
7	pp_d	photo period as a function of latitude and time of the year	hours	(13)
8	pk	photo periodic constant	-	(13)
9	pthr	photo periodic threshold	hours	(13)
10	SW_o	soil moisture constant as a fraction of the water index	%	(14)
11	Q_{min}	minimum value of quality parameter	x [a]	(17)
12	Q_{max}	maximum value of quality parameter	x	(17)
13	K_1	defines the position of the inflection point	-	(17)
14	K_2	defines the curvature of the function	-	(17)

[a] unit depends on parameter

TABLE 3

Statistics of NIRS validation

	n	m	min-max	s	SEP	RSQ
		(% of d.m.)				
crude protein	83	20,9	6,7 - 37,8	6,8	1,3	0,95
OM digestibility	93	78,1	58,2 - 88,7	5,8	2,0	0,88

(n: number of samples, m: mean, min/max: minimum/maximum, s: standard deviation, SEP: standard error of performance, RSQ: R-Square)

TABLE 4

Descriptive regression statistics of simulated and experimentally observed variables of different sward types

Variable	Sward Type	RSQ	CV	P>F
Dry matter yield	R	0.85	25.4	0.0001
	LP	0.78	33.2	0.0001
	LM	0.92	19.3	0.0001
	R x LP	0.88	24.7	0.0001
	R x LM	0.90	21.9	0.0001
Crude protein content	R	0.77	7.9	0.0001
	LP	0.83	12.0	0.0001
	LM	0.77	14.3	0.0001
	R x LP	0.72	9.1	0.0001
	R x LM	0.67	12.2	0.0001
Digestibility	R	0.82	2.3	0.0001
	LP	0.75	1.5	0.0001
	LM	0.85	2.2	0.0001
	R x LP	0.82	2.3	0.0001
	R x LM	0.71	3.3	0.0001

(R: red clover; LP: perennial ryegrass; LM: italian ryegrass; RSQ: R-Square; CV: coefficient of variation; P>F: probability)

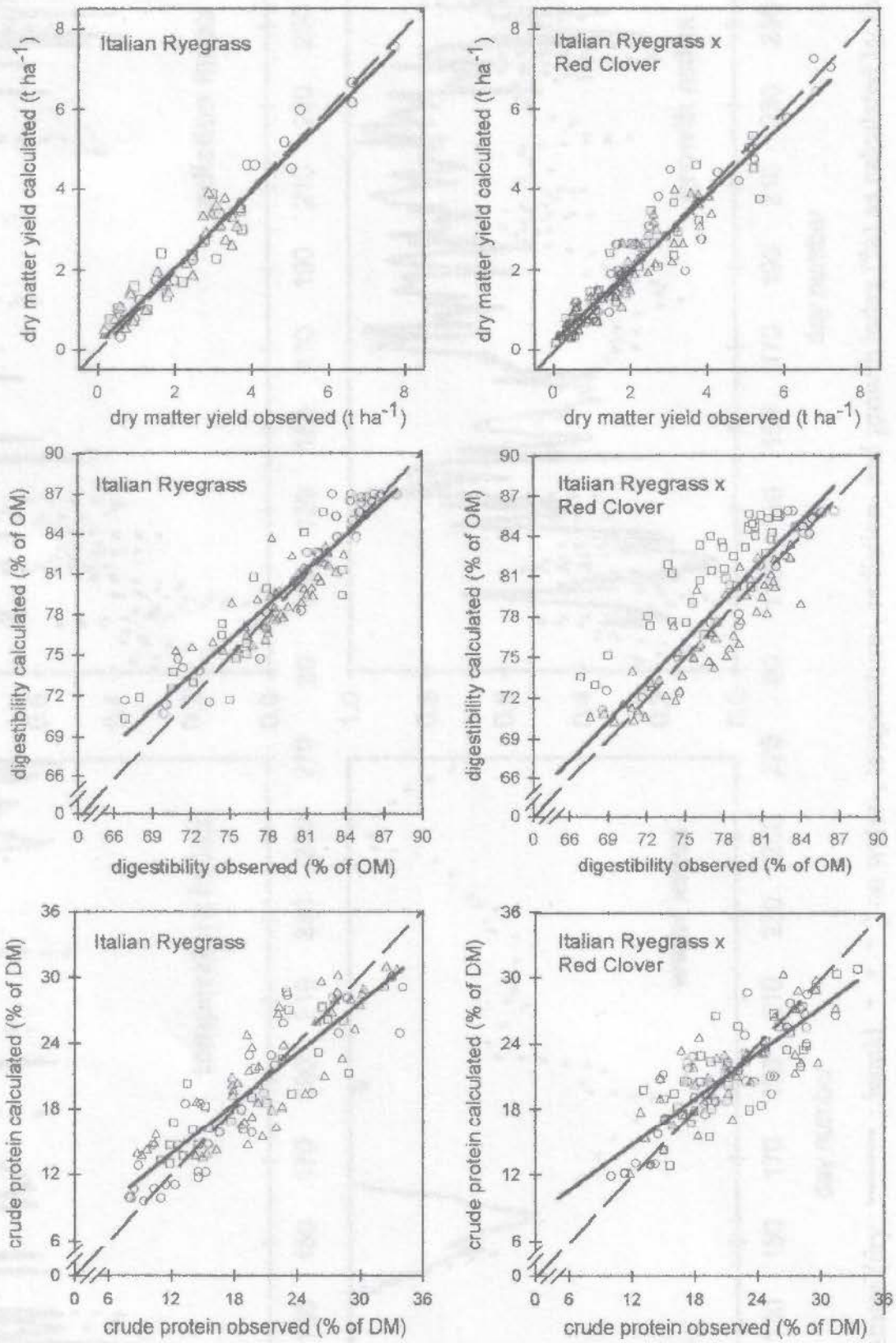

Fig.1. Observed and simulated dry matter yield, digestibility and crude protein content of Italian Ryegrass swards and Red Clover x Italian Ryegrass-mixtures (O: spring-, □: summer-, Δ: autumn-submodels; ——— : regression line; regression statistics s. TABLE 2)

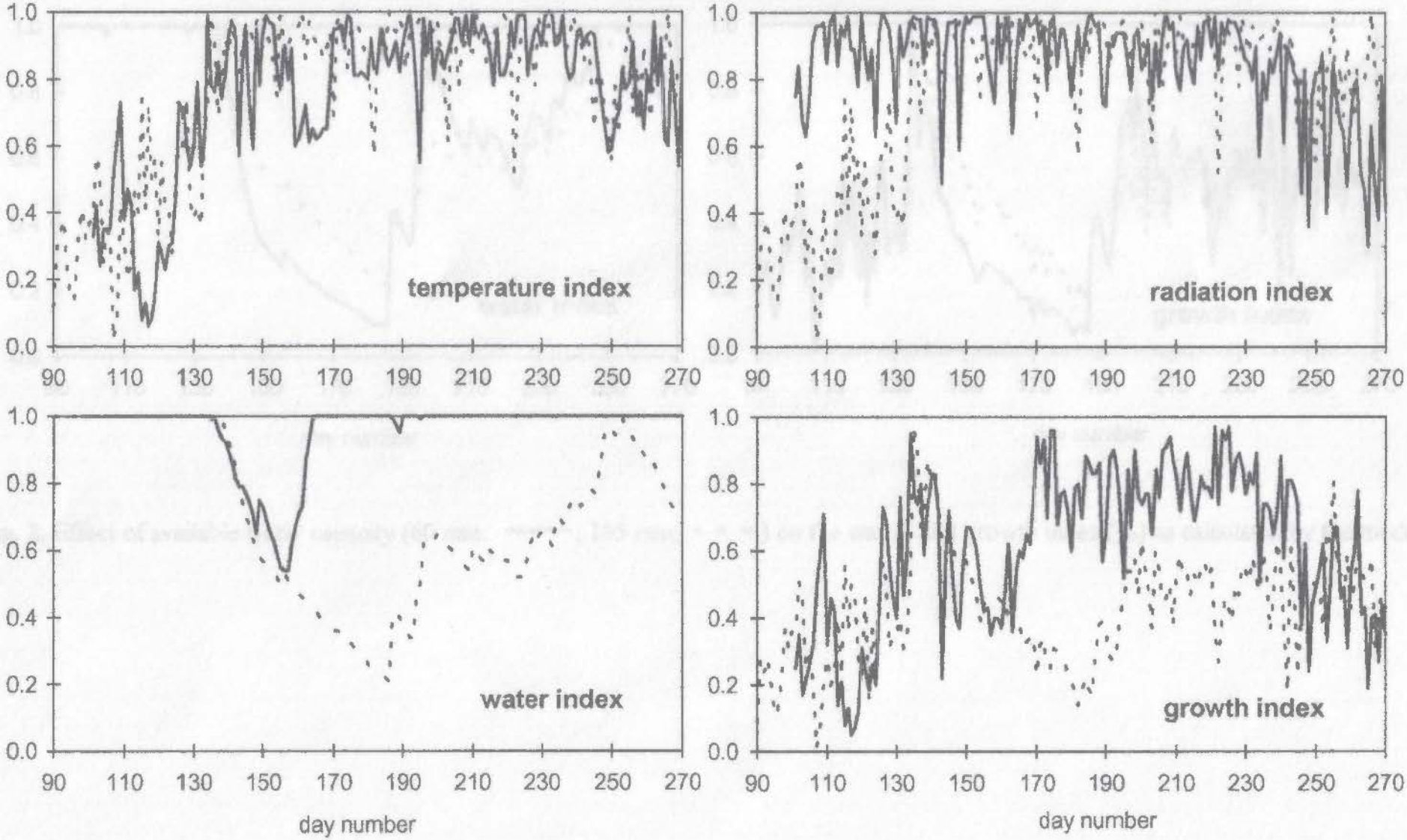

Fig. 1. Effect of climate (dry: ——— ; humid: - - -) on water-, temperature-, radiation- and growth index (%) as calculated by the model

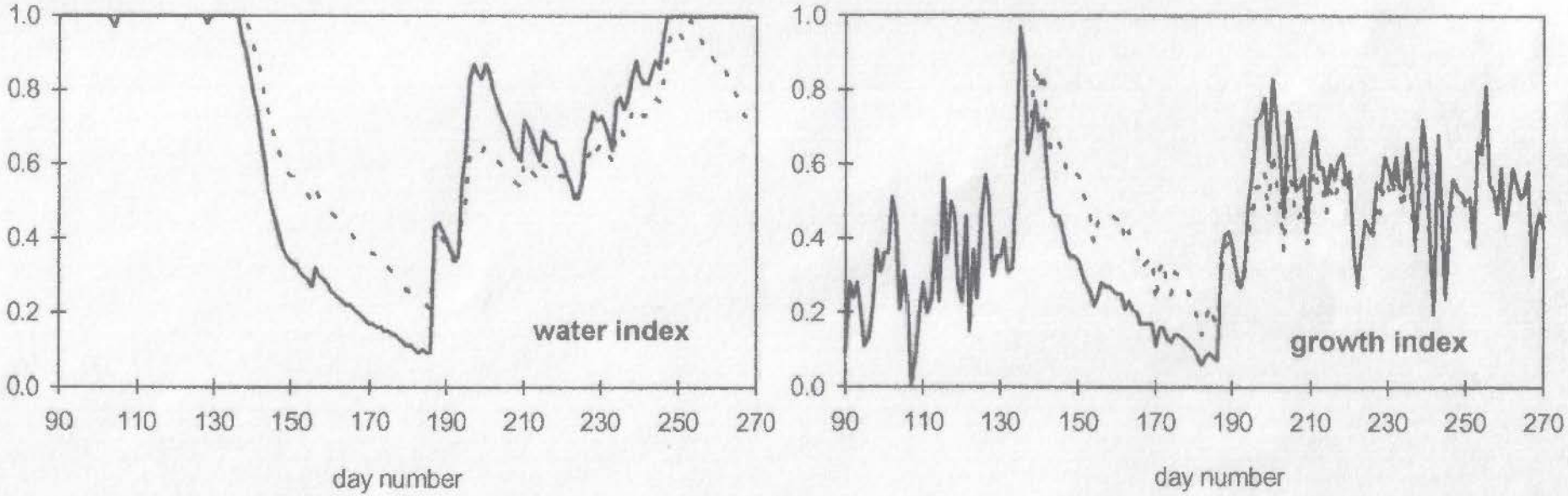

Fig. 2. Effect of available water capacity (60 mm: ——; 105 mm: - - -) on the water- and growth index (%) as calculated by the model

Simulation of Forage Production Systems with a Weather-based Model

II. Model Application - Dry Matter yield and fodder quality

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ABSTRACT

Keywords: Red clover/grass-mixtures, dynamic model, yield, quality.

INTRODUCTION

MATERIALS AND METHODS

The choice of harvest date determines yield and quality of fodder crops. In the model one critical variable (e.g. DM yield) can be used to control harvest date as well as several critical variables simultanously (DM yield and OM digestibility). In the simulations presented in this paper three critical values (DM yield, OM digestibility and CP content) are used simultanously for different cutting regimes. This means that in a 3cut-system a new growth periode starts if the first cut yields more than 5 t DM/ha rsp. if the aftercut yields more than 4 t DM/ha or if the digestibility rsp. protein content in the herbage biomass reaches the 73% rsp. 12% level (Table 1).

TABLE 1

In order to demonstrate the effects of atmospheric conditions on the yield and quality development of different swards two sets of wheather datas were used in simulations

which represent extremly contrary growth conditions (Table 2). The year 1985 represents a cool and humid climate in spring and summer, whereas spring 1992 started early and summer was hot and dry.

TABLE 2

RESULTS AND DISCUSSION

Patterns of yield formation as predicted by the model are strongly affected by species (Fig 2). Comparing to the clover/grass-mixture the pure stand of Italian Ryegrass shows a rapid development in early spring which corresponds to the low temperature demands of this species, particularly by considering the intensive fertilization of the pure grass sward. Both swards suffer water deficencies in the second cut with the dry climate, but the mixture profits better by high rainfall in summer. With the onset of precipitation in autumn and decreasing evaporation the grass sward recovers better than the unfertilized mixture.

As experiences in farming practise and several field trials show, the demands for fodder quality normally determine harvest dates in intensive fodder production in north-western europe. The accelerated decline of digestibility of both swards under dry conditions point at an enhanced phenological development of plants as descriped by Minson (19??). The effect

CONCLUSIONS

ACKNOWLEDGEMENTS

REFERENCES

TABLE 1

Thresholds of yield and quality of swards to release a harvest in the simulations

Cutting frequency	DM yield (t/ha)					OM digestibility (% d. OM)	CP content (% d. TM)
3 cuts p.a.	5	4	4			> 73	> 12
4 cuts p.a.	4	3	2.5	2.5		> 76	> 12
5 cuts p.a.	2.5	2	2	2	2	> 78	> 12

(DM: dry matter; OM: organic matter; CP: crude protein)

TABLE 2

Average temperature and precipitation at Kiel-Holtenau in the spring and summer of 1985 and 1992

	April	Mai	Jun	July	Aug.	Sept.
Temperature (°C)						
1985						
1992						
Precipitation (mm)						
1985						
1992						

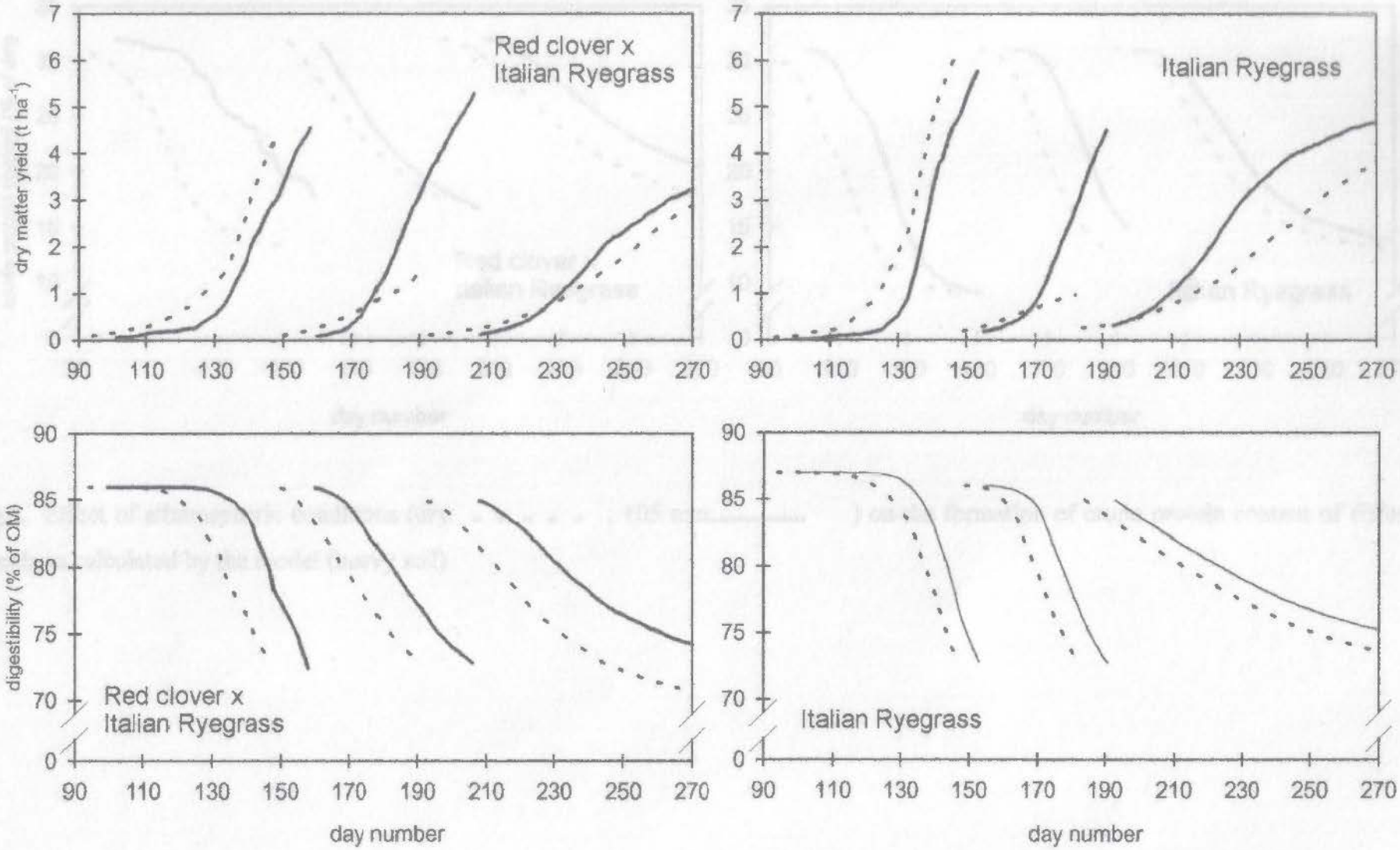

Fig.1. Effect of athmospheric conditions (dry:- - - - - ; humid: ———) on the formation of dry matter yield and digestibility of different swards as calculated by the model (heavy soil)

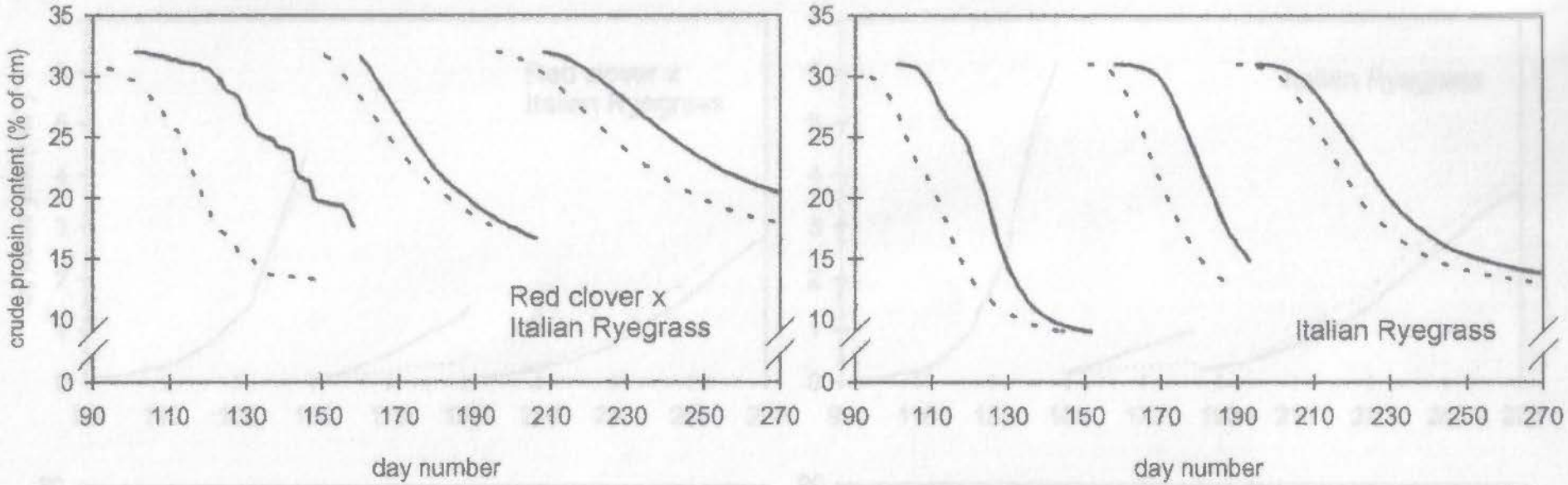

Fig.2. Effect of athmospheric conditions (dry: - - - - - ; 105 mm: ———) on the formation of crude protein content of different swards as calculated by the model (heavy soil)

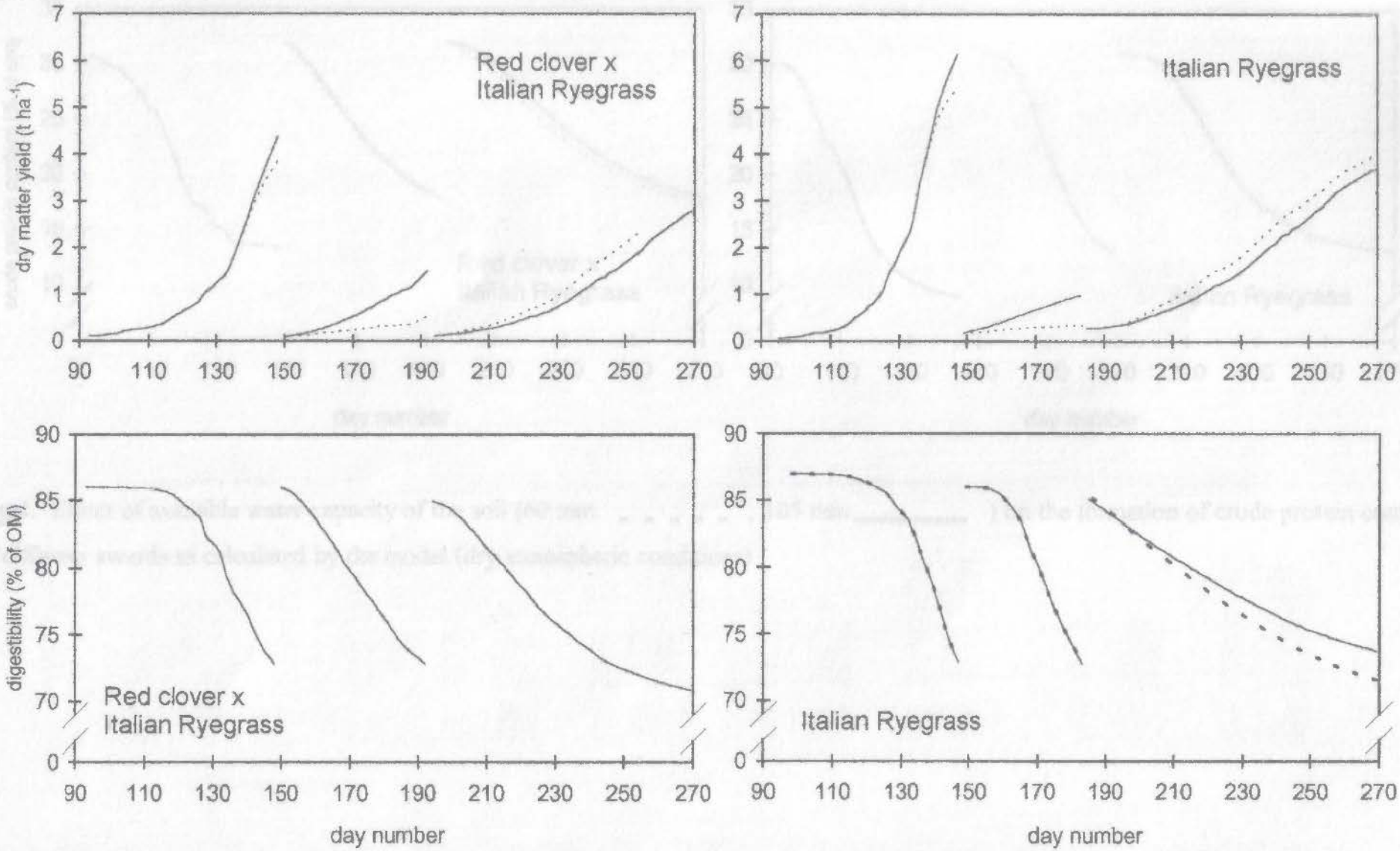

Fig.3. Effect of available water capacity of the soil (60 mm: - - - - - ; 105 mm: ————) on the formation of dry matter yield and digestibility of different swards as calculated by the model (dry atmospheric conditions)

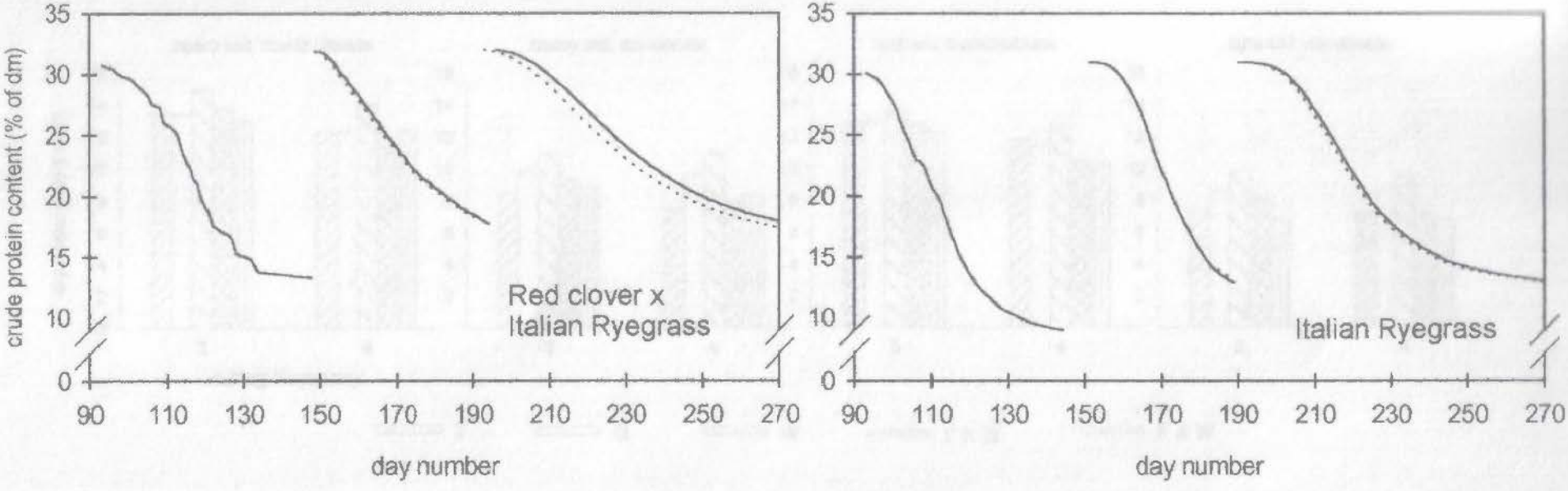

Fig.4. Effect of available water capacity of the soil (60 mm: - - - - - ; 105 mm:_____) on the formation of crude protein content of different swards as calculated by the model (dry atmospheric conditions)

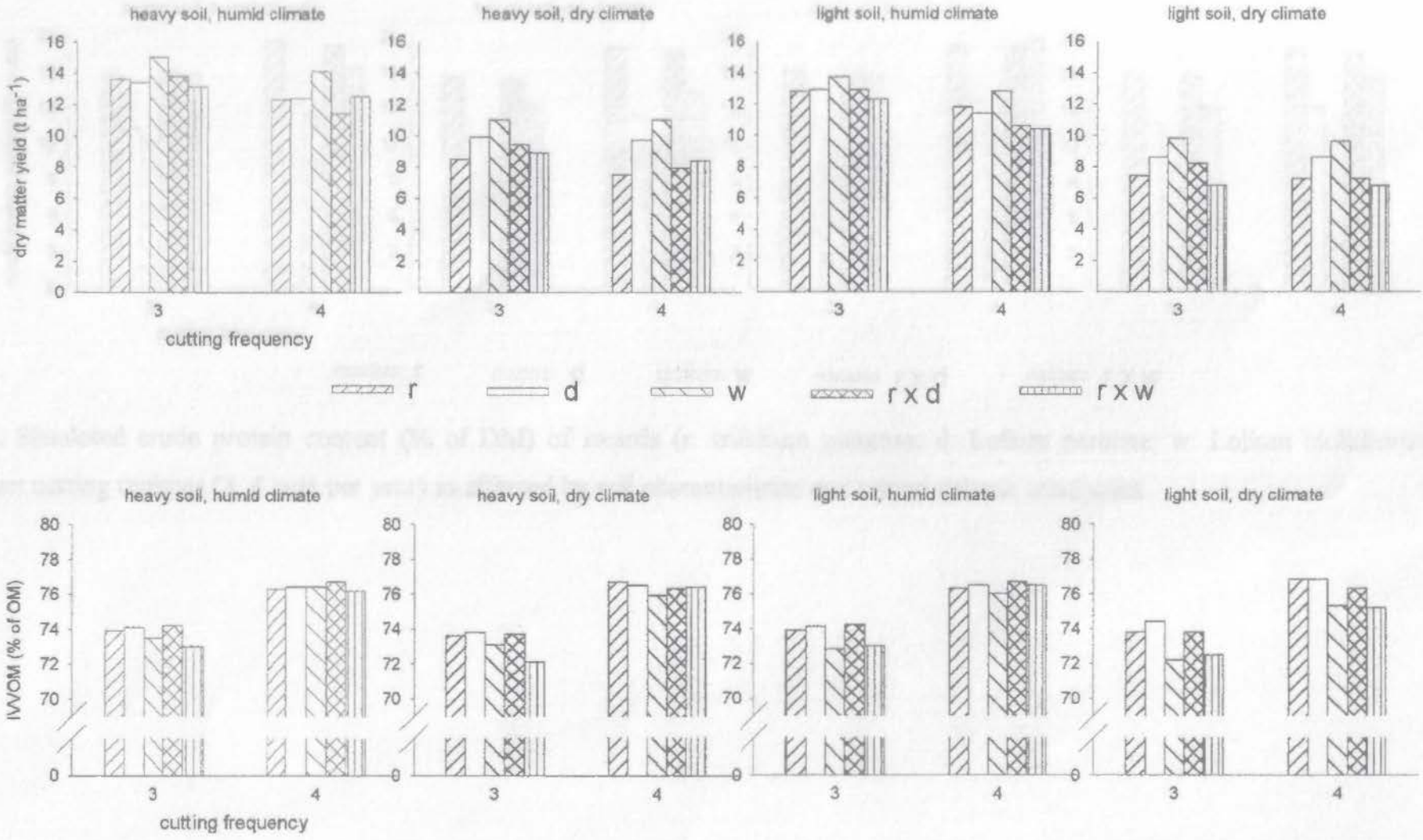

Fig. 5. Simulated dry matter yield ($t \cdot ha^{-1}$) and digestibility (% of OM) of swards (r: trifolium pratense; d: Lolium perenne; w: Lolium multiflorum) under different cutting regimes (3, 4 cuts per year) as affected by soil properties and athmospheric conditions.

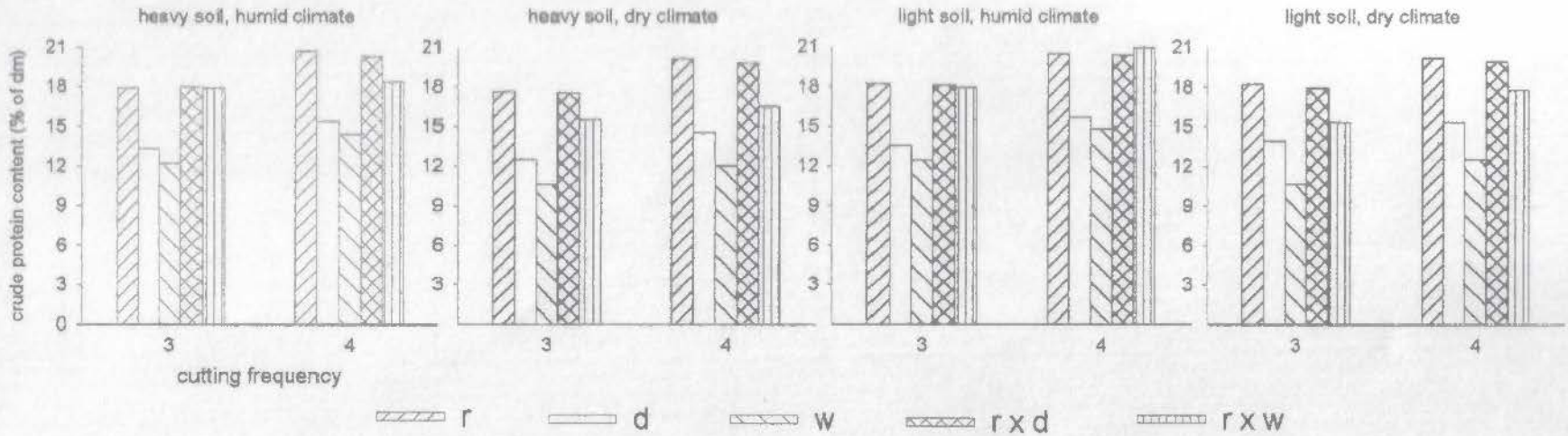

Fig. 6. Simulated crude protein content (% of DM) of swards (r: trifolium pratense; d: Lolium perenne; w: Lolium multiflorum) under different cutting regimes (3, 4 cuts per year) as affected by soil characteristics and athmo-spheric conditions.